

Xinhui CHEN

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EDUCATION

South China University of Technology

Guangzhou, CN

M.Eng. of Safety Engineering

Sep. 2022 – Present

- **Average Score:** 87 / 100
- **Core Courses:** *Applied Statistics Theory and Method (93), Fundamentals of Elasticity and Finite Element Method (87), Safety Production Technology (90)*

Northeastern University

Shenyang, CN

B.Eng. of Safety Engineering

Sep. 2018 – Jun. 2022

- **GPA:** 3.44 / 4.0 **Ranking:** 6 / 56 (Top 10.7%)
- **Core Courses:** *Control of Noise & Vibration (96), Mechanical Safety (97), Safety Measurement and Instrument (96), C Programming Language (90)*
- **Course design:** *Fire and Explosion Prevention (A), Electric Safety (A), Accident Investigation (A)*

RESEARCH INTERESTS

- Computational Fluid Dynamics (CFD)
- Fluid Mechanics
- Artificial Intelligence and Machine Learning
- Simulation, Modelling and Numerical Methods
- CO₂ Sequestration & Unconventional Oil and Gas Extraction

PUBLICATIONS

- [1] **Chen, X.**, Huang, S., Guan, T., & Fu, H. (2025). An interpretable predictive model for liquid holdup of hilly-terrain oil-gas pipelines based on machine learning and Shapley additive explanations. *ACS Omega*. DOI: 10.1021/acsomega.4c10874.
- [2] Liu, Z. (Advisor), **Chen, X.***, Tian, F.*, Zhu, W., Hu, Z., Su, W., & Wang, Z. (2023). Dynamic effects of rough elements in fractures on coal permeability during different stages of methane extraction. *Gas Science and Engineering*, 119, 205118. DOI: 10.1016/j.jgsce.2023.205118.
- [3] **Chen, X.**, Huang, S., Guan, T., Fu, H., & Pan, Z. (Accepted). Prediction model of liquid holdup of multiphase pipelines based on PSO-BP algorithm. *Chemical Engineering (China)*.
- [4] **Chen, X.***, Tang, Z., & Mo, Y. (2023). Design, Processing and Performance Testing of Extra-Large-Flow Centrifugal Model Pumps. *Journal of Physics: Conference Series*, 2610(1), 012022. DOI: 10.1088/1742-6596/2610/1/012022.
- [5] Yang, G., **Chen, X.***, Hu, Y., Huang, S., & Xu, H. (2024). Analysis of vibration characteristics of cryogenic insulated cylinders based on ANSYS. *Journal of Physics: Conference Series*, 2860(1), 012016. DOI: 10.1088/1742-6596/2860/1/012016.
- [6] Xie, H., Ou, C., **Chen, X.**, & Yang, G. (2023). Numerical analysis of the temperature rise at fast filling of hydrogen storage cylinder based on fluid-thermal-solid coupling. *Journal of Physics: Conference Series*, 2610(1), 012025. DOI: 10.1088/1742-6596/2610/1/012025.

PATENTS

- [1] Huang, S. (Advisor), **Chen, X.**, Guan, T., & Fu, H. (2024). Prediction method for liquid holdup of hilly-terrain oil-gas pipelines based on SSA-BP neural network. (Chinese Patent No. 202410626009.7)
- [2] Huang, S. (Advisor), **Chen, X.**, Hu, Y., & Yang, G. (2024). A finite element analysis method for vibration characteristics of vertical low temperature insulated cylinder system. (Chinese Patent No. 202411477988.0)

RESEARCH PROJECTS

AI-based predictive model for flow characteristic in hilly-terrain multiphase flow pipelines

Principal Investigator | Advisor: Prof. Si Huang

Jun. 2023 – Present

- Simulated the oil-gas flow in a hilly-terrain multiphase pipeline under 18,900 different operating conditions using the multiphase flow simulation software OLGA
- Developed six AI-based prediction models, utilizing a dataset of simulation results
- Established an interpretable model integrating SHapley Additive exPlanations (SHAP) to reveal the global contributions and interactions of factors in the AI black box

Influence of fracture morphology on gas migration patterns in coal

Principal Investigator | Advisor: Prof. Zhengdong Liu

Mar. 2021 – Sep. 2023

- Derived a new numerical model of rough-walled fractures that innovatively simplifies the fracture surface
- Proposed a concept of effective fracture aperture that varies nonlinearly with the CH₄ extraction and CO₂-ECBM process
- Constructed a CFD model of methane extraction using COMSOL Multiphysics, in which the governing equations for gas diffusion and seepage influenced by the new numerical model were imported
- Analyzed the dynamic effects rough element and operating conditions on CH₄ extraction and CO₂-ECBM

Simulation and testing of vibration characteristics of cryogenic insulated cylinders

Principal Investigator | Advisor: Prof. Si Huang

May 2024 – Present

- Formulated a modal analysis model for the cryogenic insulated cylinder utilizing ANSYS
- Delved into the natural frequency trends of the cylinder across varying filling percentages

Fluid-thermal-solid coupling of carbon fiber full-wrapped hydrogen cylinders during fast inflating

Co-investigator | Advisor: Prof. Si Huang

Jan. 2023 – Jun. 2023

- Established a fluid-thermal-solid coupled CFD model for fast inflating of carbon fiber full-wrapped hydrogen cylinders based on ANSYS
- Verified that the maximum stress point on the cylinder corresponds precisely to the vulnerability pinpointed by acoustic emission signal mapping

Hydraulic model design and test of large capacity vertical centrifugal pump

Co-investigator | Advisor: Prof. Si Huang

Sep. 2022 – Jan. 2023

- Designed a model pump based on a prototype pump according to similarity principles
- Constructed physical models of centrifugal pumps utilizing CFTurbo
- Established the CFD model of centrifugal pumps using CFX

HONORS & AWARDS

South China University of Technology Graduate Scholarship (three times, RMB18,000)

Sep. 2022 – Jun. 2025

Merit Undergraduate Thesis, Northeastern University (Top 2%)

Jun. 2022

Likang Group Safety Engineering Scholarship (RMB 3,000)

Oct. 2021

The National Undergraduate Mathematics Competitions, Third prize of national

Dec. 2020

The National Undergraduate Mathematics Competitions, Second prize of provincial

Dec. 2020

Northeastern University Merit Student Scholarship, Third prize (four times)

Sep. 2018 – Jun. 2022

ACADEMIC CONFERENCES

The 4th International Conference on Fluid and Chemical Engineering, Wuhan, China

July 26 – 27 2024

The 3th International Conference on Fluid and Chemical Engineering, Wuhan, China

July 21 – 22 2023

SKILLS

Technology: ANSYS, COMSOL Multiphysics, OLGA, MATLAB, SpaceClaim

Language: Mandarin (Native), English (IELTS: 6.5), Cantonese (Basic)